Smaller eleven variable consistency equations

Research supplement for ProveIt

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1 Eleven positive unknowns with denominator reduced absorption

The new equation has eleven positive unknowns, exact degree 958, and 117 charged binary arithmetic operations. A second equation trades four degrees for two operations: degree 962 and 115 operations. The full carrier's two-equation system has degrees $(2, 956)$ and uses 112 operations after the standard positive transport shift. These improve both pinned eleven-positive frontiers, 1154/123 and 1140/129. They do not dominate the twelve-positive degree-901 construction. The argument uses no Sun representation theorem.

Two changes make this possible. First, the multiplier-one norm from source44 remains sound without the deleted index equation: its main-root bound needs $d(I\sqrt{G} - c\sqrt{D}) > b + c\sqrt{D}$, not the stronger $I\sqrt{G} - c\sqrt{D} > 1$. Second, quotient absorption needs denominator Y, not $E = XY$. The short Pell-index argument recovers the full E-divisibility after the weaker Y-divisibility has done its job.

1.1 Explicit equation

Take precisely the positive integer unknowns

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t.$$

Fix the complete authentic normalized blank-K215 coefficient tuple (β, K_0, M_C, M_F), with M_F already the shifted high mask. The delivered fixed instance is the materialized Wang943 121×8801 tuple with $\beta = 2^{1064921} - 1$; its four payloads and hashes are included. The proof is uniform in any whole authentic tuple. Set

$$\begin{gathered}
q = \beta J + 1, \quad s_0 = s + 3q, \quad X = wq, \quad Y = s_0 q^3, \quad E = XY, \\
k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \\
\Delta = a^2 + H = (a+2)^2 - 1, \quad \ell = EkY, \quad A = 1 + \ell(\ell + k), \\
G = 1 + \Delta c^2, \quad b_0 = X + ac, \quad C_w = q - F - \alpha, \\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
N_t = (K_0 + w)C_w + q - F - z(q-1), \\
I = ic^2, \quad D = 1 + \Delta I^2, \quad Q = \Delta(D-1) = (\Delta I)^2, \\
d = Q - 1, \quad v = cT, \quad b = c + RD, \\
e = Q(Dv^2 + b^2) - 1, \quad B_* = 2Qvb, \quad K = IT + 1, \\
L_0 = 1 + HK, \quad S = HY, \quad L = YL_0.
\end{gathered}$$

The computed register K is not the fixed coefficient K_0. For the full carrier put $\delta = (a+2)c - \ell$; for the shorter one put $\delta = ac - \ell$. In either case define

$$W = Y\big[b_0 + H(t + b + \delta)\big] + H(k - R - 1)$$

and the integer polynomial

$$N = \prod_{\substack{g=\pm\sqrt{G} \\ \tau=\pm\sqrt{A} \\ f=\pm\sqrt{D}}} \Big[d(W - Lg - S\tau)^2 - S^2 e + S^2 B_* f \Big]. \tag{1}$$

The formal signs define a conjugate product, not independently assumed Galois automorphisms. Every other symbol is a polynomial abbreviation, not another unknown. The proposed single equation is

$$\boxed{N_t(1+N) - 1 = 0}. \tag{2}$$

The two-equation presentation is $N_t = 1$, $N = 0$.

For an explicit division-free scalar evaluator, set

$$\begin{aligned}
d_n &= dS^2 A, & H_Q &= S^2 Q, & H_P &= H_Q D v^2, & H_U &= H_Q b^2, \\
H_e &= H_P + H_U - S^2, & j &= 4 H_P H_U, & & & & \\
\mathcal{T} &= 2d(W^2 + L^2 G), & \mathcal{K} &= [d(W^2 - L^2 G)]^2, & & & & \\
A_* &= 2(d_n + H_e), & C_* &= (d_n - H_e)^2 - j, & & & & \\
u &= \mathcal{T} - A_*, & r &= C_* - \mathcal{K}, & c_* &= 16 d_n j, & & \\
V_* &= c_* - u(u\mathcal{T} + 2r), & U_* &= r^2 - u^2 \mathcal{K}, & & & & \\
N &= \mathcal{K} V_*^2 - U_*(\mathcal{T} V_* - U_*). & & & & & &
\end{aligned}$$

This is a universal polynomial identity, including degenerate radicands. The emitted JSON circuits expand every square and count every binary gate.

1.2 Weighted main-root separation

Let $x_0 = d(W^2 + L^2 G)$. Since $\mathcal{T} = 2x_0$ and $\mathcal{K} \equiv x_0^2 \pmod 4$, the norm obeys $N \equiv (U_* - x_0 V_*)^2 \pmod 4$. Thus a root of (2) forces $N = 0$, $N_t = 1$: the alternative integer unit $1 + N = -1$ would make $N = -2$. The inherited transport interface then gives

$$q \geq 16, \quad C_w > 0, \quad 3q < R < q^4 - 2.$$

Our positive shift implies

$$X \geq 16, \quad Y > 3q^4 > 3R, \quad E > R + 1, \quad k \geq 2, \quad kY < c < 2kY, \quad c > 96, \quad a + 2 < 2E.$$

No assertion $k > E$ has been used.

For positive real $g = \sqrt{G}$, $f = \sqrt{D}$, $\tau = \sqrt{A}$, direct comparison yields

$$1 < A < \Delta c^2 \leq \Delta I = \sqrt{Q}, \qquad g > \tau, \quad g < 2Ec, \quad f < 2EI.$$

The exact identity $(Ig)^2 - (cf)^2 = I^2 - c^2$ and $I \geq c^2$ give

$$Ig - cf > \frac{I}{8Ec}.$$

Moreover $b + cf < 6E^3I^2$ and $d > \frac{1}{2}E^4I^2$, so

$$d(Ig - cf) > \frac{E^3I^3}{16c} \geq \frac{E^3I^2c}{16} > 6E^3I^2 > b + cf. \tag{3}$$

This is the new estimate that removes dependence on the missing index quotient.

Every root of the inner norm has a coherent representation

$$x = \tau' + y, \quad (\tau')^2 = A, \quad (f')^2 = D, \quad dy^2 = Q(vf' - b)^2 - 1.$$

These roots may be complex. With $Z = b + vf > 1$,

$$|y| < Z + Z/d, \qquad Z/d \leq T(b + cf)/d < T(Ig - cf)$$

by (3). Hence $|x| < b + (IT + 1)g = b + Kg$. But the negative main sign gives

$$\frac{W + Lg}{S} = b + t + \delta + \frac{k - R - 1}{Y} + \frac{b_0}{H} + \left(K + \frac{1}{H}\right) g > b + Kg,$$

since $t + (k - R - 1)/Y > 0$. This is impossible. Quadratic conjugation therefore forces the main root g to be a positive integer, and $x = (W - Lg)/S$ is rational.

1.3 Integral inner roots

Set $C(x) = d(x^2 + A) - e$. The inner norm is

$$F(x) = \left[C(x)^2 + DB_*^2 - 4d^2x^2A\right]^2 - 4C(x)^2DB_*^2.$$

If $C(x) = F(x) = 0$, substitution gives $d^2A^2 - deA + Q^2v^2b^2D = 0$, so $Q \mid A(A - 1)$, impossible because $1 < A < \sqrt{Q}$. Also $F(0) = 0$ would imply $Q \mid (A - 1)^2$, again impossible. The rational expressions

$$f' = -\frac{C(x)^2 + DB_*^2 - 4d^2x^2A}{2C(x)B_*}, \qquad \tau' = \frac{C(x) + B_*f'}{2dx}, \qquad y = x - \tau'$$

therefore have justified denominators and satisfy $(f')^2 = D$, $(\tau')^2 = A$ and the auxiliary equation. Thus f', τ' are integers. The inherited rational Pell saturation lemma,

$$m^2V^2 - (m^2 - 1)y^2 = 1, \quad m, V \in \mathbb{Z},\ m \geq 2, \quad y \in \mathbb{Q} \quad \Longrightarrow \quad y \in \mathbb{Z},$$

with $m = \Delta I$ and $V = vf' - b$ makes y, hence x, integral. A proof of the lemma is included in the companion proof file; it uses abscissa and ordinate divisibility in the integer-coordinate Pell group. Reducing $W = Lg + Sx$ modulo Y now yields

$$Y \mid H(k - R - 1). \tag{4}$$

1.4 Short indices restore full divisibility

Write $A_0 = a + 2$ and let χ_{A_0}, ψ_{A_0} be its Pell sequences. The main and strong roots give positive indices p, m with

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \quad |f'| = \chi_{A_0}(m), \quad I = \psi_{A_0}(m).$$

Strong divisibility and $I = ic^2$ give $p \mid m$; composition modulo c then gives $c \mid m/p$, so $m \geq pc > 2p$. Also $p < c/2$ and $R < c/2$. Source38's signed auxiliary-index argument, applied before (4), yields $R \equiv \pm p \pmod{c}$, hence $p = R$. For clarity, it classifies the auxiliary Pell solution at ΔI with odd index j_0. Reducing its odd-index polynomial modulo $|f'|$ gives $j_0 \equiv \pm p \pmod{m}$, while reduction modulo c gives $j_0 \equiv \pm R \pmod{c}$. The full recurrence argument is reproduced in the companion proof file, including the signed branches.

The square A has fundamental integer-coordinate unit $(2EY + 1, 2)$, so for some $n \geq 1$,

$$|\tau'| = \chi_{2EY+1}(n), \qquad k = 2\psi_{2EY+1}(n).$$

Monotonicity and $2EY + 1 > A_0$, $c > k$ imply $n < p = R$. Since $2EY + 1 \equiv 1 \pmod{E}$,

$$k \equiv 2n \pmod{E}.$$

Now $\gcd(Y, H) = \gcd(Y, 3) \leq 3$. By (4),

$$2n \equiv R + 1 \pmod{\mu}, \qquad \mu = Y/\gcd(Y, H) > R.$$

But $|2n - R - 1| < R$, so $2n = R + 1$. It follows that $E \mid k - R - 1$ after all, and

$$h = (k - R - 1)/E > 0.$$

Dividing the carrier by Y gives

$$b_0 + H(t + b + \delta + hX) = L_0 g + H(\tau' + y).$$

Thus $g \equiv b_0 \pmod{H}$. The recovered $\gamma = (g - b_0)/H$ is positive: $n \geq 2$ gives $k > X$, hence $c > X$, and $g > (a + 1)c$. All normalized compiler equations are restored. Absolute values of τ', y are harmless; simultaneous sign reversal of f', T is covered by the inherited signed-quotient interface. This proves soundness through the calibrated blank-K215 halt equivalence.

1.5 Completeness and exact degree

For a canonical compiler witness, $X = 2^R$, $Y > 2^R$, and $R > 3q$ ensure $s_{\text{old}} = Y/q^3 > 3q$. Thus $s = s_{\text{old}} - 3q > 0$. Choose positive canonical roots and define

$$t = \gamma + Kg + \tau + y - b - \delta - hX.$$

Since $0 < \delta < g$, $y \geq b - vf$, and $\tau > \ell = EkY > k/Y > hX$, one has

$$t > \gamma + \tau + T(Ig - cf) - hX > 0.$$

This produces the coherent zero factor and proves completeness. The transport substitution $z_{\text{old}} = z + 1$ replaces $N_t = 1$ by $(K_0 + w)C_w = F + z(q - 1)$ and saves one operation; canonical witnesses have $z_{\text{old}} \geq 2$.

Let N_{12} be the earlier multiplier-one norm. The exact localized identity is

$$N = Y^{16} N_{12}\left(s_{\text{old}} = s + 3q, t_{\text{old}} = t + \frac{k - R - 1}{Y}\right).$$

Each of the eight factors contributes Y^2, and the t_{old} degree is at most 16, so denominators clear. The rational shift has degree 0, below the degree-1 supplied carrier. It cannot change the highest

form. The invertible highest-linear substitution $s_{\mathrm{old}} = s + 3\beta J$ preserves its nonvanishing. More explicitly, the rational carrier perturbation changes an unscaled factor only in degrees at most 96 and 58, below its smallest established factor degree 97. The earlier exact norm degrees 892 and 896 consequently become 956 and 960. The transport factor has exact degree 2, giving 958 and 962.

The full highest form is Y_0^{16} times

$$-2^{14} H_0^{16} d_0^8 B^8 V^4 \ell_0^2 k_0 Y_0^2 \left(2t + k_0 - \frac{iTc_0}{E_0}\right),$$

where

$$q_0 = \beta J, \quad Y_0 = (s + 3q_0)q_0^3, \quad X_0 = wq_0, \quad E_0 = X_0 Y_0, \quad c_0 = k_0 Y_0, \quad k_0 = \eta + \zeta,$$
$$I_0 = ic_0^2, \quad H_0 = 4E_0, \quad \ell_0 = E_0 c_0, \quad d_0 = E_0^4 I_0^2,$$
$$B = q_0^3 (q_0 - F) E_0^2 I_0^2, \qquad V = I_0 T \ell_0.$$

Its t coefficient is nonzero for every authentic $\beta > 0$; the displayed denominator cancels. Multiplying by $w(q_0 - F - \alpha) - zq_0$ gives the single equation's highest form. The companion proof gives the short carrier's form as well.

1.6 Verification and logical boundary

The included builder audits all gates and coefficient hashes. The independent checker passes 24 exact and 60 modular formal conjugate products per carrier, three complete finite-field polynomial rays with both illustrative and actual fixed coefficients, all stated leading forms, the transport shift, and 288 pre-index bound tests. The tests include $k = 2$ and arbitrarily large X, where the old unweighted gap shortcut is unavailable. Both normal Python and optimized Python pass. Finite checks validate the implementation, not the unbounded compiler.

The baseline is ProveIt commit `124c07c0b77a9e20e84676514d8b1ff78536ea3f`, specifically source38's index reconstruction and source44's multiplier-one norm. Complete source pins, copied proof dependencies, payloads, circuits, and the detailed mathematical proof accompany this section. The resulting theorem is that ZFC proves the absence of positive solutions equivalent to Con(ZFC). No consistency decision, global minimality theorem, independent human review, or new kernel-checked end-to-end certificate is asserted.